

HANEY KANG

Systems & Security Engineer

Ph.D. in Electrical Engineering, Korea Advanced Institute of Science and Technology (KAIST) · 2026

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Professional Summary

I'm a systems and security engineer with a Ph.D. in Electrical Engineering from KAIST. I have experience designing, implementing, and evaluating security mechanisms for Linux and cloud-native environments. My projects focus on eBPF-based runtime analysis, Seccomp policy generation, container network policy verification, and LLM-assisted security automation. As an engineer, I'm highly passionate about turning ideas into functioning prototypes and demonstrating improvements through controlled experiments.

Technical Skills

Programming: Python, C, Java, Bash

Systems & Container Security: Linux, Docker, Kubernetes, Seccomp, eBPF/BCC

Binary Analysis: IDA Pro, Capstone

Networking: Software-Defined Networking (SDN), Cilium

Methods: LLM Fine-Tuning, Runtime Analysis, Graph-Based Verification, SMT Solving (Yices2)

Research & Engineering Experience

- **Kernel-level Runtime Instrumentation**: Analyzed Linux kernel code to identify suitable eBPF instrumentation points to inspect system-call invocations and capability checks. Implemented event handling algorithms using kernel function arguments and eBPF data structures.
- **Binary & Application Analysis**: Analyzed application binaries and dynamically linked shared libraries using IDA Pro and Capstone to construct call graphs and identify reachable system calls. Used stack traces to resolve ambiguities caused by indirect calls and jumps.
- **Container Infrastructure & Security**: Managed both manual and automated deployments of containers across a broad set of Docker Official Images, applying Seccomp policies through Docker CLI commands, Docker Compose files, and Kubernetes manifests for large-scale experiments.
- **LLM-assisted Security Automation**: Applied prompt engineering and fine-tuning to adapt LLMs for container security policy generation.
- **Formal Modeling & Verification**: Modeled container network policies as formal constraints, then applied SMT solvers to detect conflicting allow/deny rules across overlapping container selectors and L4 traffic specifications.
- **Software & Pipeline Engineering**: Implemented and integrated system components into end-to-end pipelines using C, Java, and Python.

Selected Engineering Projects

BeaCon — Environment-Aware Dynamic Policy Generation. Designed an eBPF-based system that profiles containers across representative runtime configurations and workloads, then generates tunable least-privilege Seccomp policies. Among 54 images compatible with all comparison tools, about 80% received policies allowing 100+ fewer syscalls than Confine under the functionality-maximized setting.

Prompt2Seccomp — Semantics-Aware Seccomp Policy Generation. Developed a pipeline combining container metadata, application documentation, fine-tuned LLM inference, and sandboxed runtime validation. On 95 evaluation images, achieved a precision of 0.6621 and an F1 score of 0.7332, compared with 0.3334 and 0.4788 for Confine. Manuscript in preparation.

VeriKube — Container Network Policy Verification. Designed a graph-based system that translates Kubernetes/Cilium network policies into formal constraints and uses an SMT solver to detect policy conflicts. Achieved at least 7× higher memory efficiency and up to 20,000× faster verification over the evaluated baseline.

Archon — Requirement-Guided Runtime Evidence Generation. Explored an approach that connects high-level deployment requirements to syscall-level evidence through binary analysis, behavior activation, and runtime tracing.

Selected Publications

Haney Kang, Eduard Marin, Myoungsung You, Diego Perino, Seungwon Shin, and Jinwoo Kim (2026). "BeaCon: Automatic Container Policy Generation Using Environment-Aware Dynamic Analysis."
Computers & Security, Vol. 162, Article 104789.

Haney Kang and Seungwon Shin (2022). "VeriKube: Automatic and Efficient Verification for Container Network Policies."
IEICE Transactions on Information and Systems, Vol. E105-D, No. 12, pp. 2131–2134.

Haney Kang, Jinwoo Kim, and Seungwon Shin (2021). "MiniCon: Automatic Enforcement of a Minimal Capability Set for Security-Enhanced Containers."
IEEE International IoT, Electronics and Mechatronics Conference (IEMTRONICS).

Manuscript in Preparation

Haney Kang, Myoungsung You, Seungwon Shin, and Jinwoo Kim. "Prompt2Seccomp: Semantics-Aware Automated Security Policy Generation for Containers Using Large Language Models."

Status: Manuscript in preparation; not yet submitted.

Education

Ph.D. in Electrical Engineering, KAIST Mar. 2020 – Aug. 2026

M.S. in Electrical Engineering, KAIST Mar. 2018 – Feb. 2020

B.S. in Electrical Engineering, KAIST Mar. 2013 – Feb. 2018

Additional Experience

KAIST Teaching Assistant 2018 – 2022
Supported courses in Programming Structure, Advanced Programming Techniques, and Network Systems and Security.

KAIST Undergraduate Research Intern 2017 – 2018
Participated in research on software-defined networking security frameworks, and network systems security.